

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination–Summer 2024 Course: B. Tech. Branch : Mechanical Engineering and Allied branches Semester: V Subject Code & Name: Renewable Energy Sources BTMOE505B Max Marks: 60 Date: 10/ 07/ 2024 Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome(CO)on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q.1	Solve Any Two of the following.		
A)	Enlist the “Renewable Energy Sources” and explain importance in current scenario.	CO1	6
B)	Explain the term ‘Pyrheliometer’ and ‘Pyranometer’. With neat sketch explain construction and working of any one ‘Pyrheliometer’.	CO1	6
C)	With neat sketch explain the ‘structure of sun and solar radiation’ on earth.	CO1	6
Q.2	Solve Any Two of the following.		
A)	Explain advantages and limitations of ‘Concentric collectors’ over - ‘Flat plate collector’.	CO2	6
B)	State and explain different types of ‘Concentric collector’ with their-applications.	CO2	6
C)	Explain construction and selection criteria for ‘Flat plate collector’.	CO2	6
Q.3	Solve Any Two of the following.		
A)	With neat sketches explain construction, working and applications - of ‘Solar Still’.	CO3	6
B)	With neat sketches explain construction, working and applications of different types of ‘Solar dryer’.	CO3	6
C)	With neat sketches explain construction and working of ‘Photo Voltaic cell’ with their applications.	CO3	6
Q.4	Solve Any Two of the following.		
A)	With neat sketches explain construction and working of ‘Wind Mill’.	CO4	6
B)	What is ‘Biomass energy? Explain with neat sketch any one type of - ‘Bio gas plant’.	CO 4	6
C)	Explain the advantages and limitations of ‘Wind Energy Sources’.	CO4	6
Q.5	Solve Any Two of the following.		
A)	With neat sketch explain principle of ‘Tidal energy’ and explain construction and working of ‘Tidal power plant’.	CO4	6
B)	Explain the principle of ‘OTEC’ and with neat sketch explain construction and working of OTEC plant.	CO4	6
C)	State principle and with sketch explain ‘Hydro-electric power plant’.	CO4	6
End			